

Nyquist

Write a function called **IsNyquistSatisfied**. The goal of the function is to confirm a proposed sampling rate, **F_s**, is high enough to avoid aliasing. The function takes as input **fmax** and **F_s**, and it outputs a logical value of true if aliasing is avoided. The second output, **F_{s_min}**, contains the Nyquist sampling rate.

Function ?

 Save Reset MATLAB Documentation (<https://www.mathworks.com/help/>)

```
1 function [y,Fs_min] = IsNyquistSatisfied(fmax,Fs)
2 niquist_rate = 2*fmax;
3 y = Fs > niquist_rate;
4 Fs_min = niquist_rate;
5 end
```

Code to call your function ?

 Reset

```
1 fmax = 16000;
2 Fs = 28800;
3 [y,Fs_min] = IsNyquistSatisfied(fmax,Fs) % false, with Fs_min=32000
4
5 fmax = 16000;
6 Fs = 44100;
7 [y,Fs_min] = IsNyquistSatisfied(fmax,Fs) % true, with Fs_min=32000
```

 Run Function

?

Assessment: All Tests Passed

Submit

?

 Sampling fast enough Sampling not fast enough Sampling almost fast enough, but not quite